Parameter-Free Prediction of Superconductivity

Anjali Mishra PH22B006

Department of Physics, Indian Institute of Technology Madras, Chennai, 600036, Tamil Nadu, India.
Prof. Shantanu Mukherjee, PH5680 (Superconductivity and its Applications).

Abstract

This term paper presents an overview of the superconducting density functional theory developed by Lüders *et al.*, which extends conventional density functional theory to superconducting systems. The formalism introduces a multicomponent framework based on the electronic density, anomalous density,and nuclear density matrix,enabling a fully *ab initio* description of superconductivity.The theoretical model, self-consistent Kohn–Sham equations,and superconducting gap equation are discussed in detail. The analysis highlights how the theory incorporates electron-electron and electron-phonon interactions on equal footing while reproducing the essential features of BCS and Eliashberg theory without relying on empirical parameters.This approach provides a predictive first-principles framework for studying conventional superconducting materials.

1 Introduction

Superconductivity is one of the most remarkable quantum phenomena in condensed matter physics,characterized by zero electrical resistance and the complete expulsion of magnetic fields below a critical temperature.Since its discovery in 1911,understanding the microscopic origin of superconductivity has remained one of the central challenges of theoretical physics.

A major breakthrough came with the Bardeen–Cooper-Schrieffer (BCS) theory,which explained superconductivity as the formation of Cooper pairs due to an effective attractive interaction between electrons mediated by lattice vibrations [2]. The BCS theory successfully describes many universal properties of conventional superconductors,including the existence of an energy gap and the relation between the superconducting gap and the critical temperature.However,it does not provide accurate material-specific predictions, particularly for strongly coupled superconductors such as niobium and lead.

To overcome these limitations,Eliashberg developed a more advanced theory that incorporates retardation effects arising from electron-phonon interactions [3]. Eliashberg theory significantly improved the description of strong-coupling superconductors,but its practical implementation requires the Coulomb pseudopotential parameter μ^*, which is usually treated as an empirical quantity. This limits its predictive capability as a fully first-principles theory.

Density Functional Theory (DFT),introduced by Hohenberg and Kohn [4] and later reformulated in a practical computational framework by Kohn and Sham [5],has become one of the most successful methods for describing the electronic properties of condensed matter systems.However, conventional DFT cannot directly describe superconductivity because it does not explicitly include the superconducting order parameter.

The work of Lüders *et al.* [1] extends the density functional framework to superconducting systems by introducing a multicomponent formalism involving the electronic density,anomalous density,and nuclear density matrix.This approach treats electron-electron and electron-phonon interactions on equal footing and eliminates empirical parameters, making it truly *ab initio* theory of superconductivity.

The objective of this term paper is to present the main ideas of this formalism, discuss its theoretical model,analyze its significance,and evaluate its importance in modern superconductivity research.

2 Model

The theoretical framework proposed by Lüders *et al.* extends conventional density functional theory to superconducting systems by introducing a multicomponent formalism.Unlike ordinary density functional theory, which depends only on the electronic density,this formalism introduces three basic densities:the electronic density,the anomalous density,and the nuclear density matrix. These quantities together provide a complete description of the superconducting electron-nuclear system.

The full many-body Hamiltonian of the system is written as

$$\hat{H} = \hat{T}_e + \hat{U}_{ee} + \hat{T}_n + \hat{U}_{nn} + \hat{U}_{en}, \tag{1}$$

where $\hat{T}_e$ is the kinetic energy operator of electrons,$\hat{U}_{ee}$ is the electron-electron Coulomb interaction, $\hat{T}_n$ is the nuclear kinetic energy, $\hat{U}_{nn}$ is the nucleus-nucleus Coulomb repulsion,and $\hat{U}_{en}$ describes electron-nucleus interaction [1].

The electronic density is defined as

$$n(\mathbf{r}) = \sum_{\sigma} \langle \hat{\psi}^{\dagger}_{\sigma}(\mathbf{r}) \hat{\psi}_{\sigma}(\mathbf{r}) \rangle, \tag{2}$$

where $\hat{\psi}^{\dagger}_{\sigma}(\mathbf{r})$ and $\hat{\psi}_{\sigma}(\mathbf{r})$ are electron creation and annihilation field operators for spin σ, and $\mathbf{r}$ is the electron position vector.This quantity describes the probability density of finding an electron at position $\mathbf{r}$.

The superconducting order parameter is represented by the anomalous density

$$\chi(\mathbf{r}, \mathbf{r}') = \langle \hat{\psi}_{\uparrow}(\mathbf{r}) \hat{\psi}_{\downarrow}(\mathbf{r}') \rangle, \tag{3}$$

which describes the pairing correlation between two electrons located at $\mathbf{r}$ and $\mathbf{r}'$.A nonzero value of χ indicates Cooper-pair formation and therefore superconductivity.

The nuclear degrees of freedom are described by the nuclear density matrix

$$\Gamma(\underline{R}) = \langle \hat{\Gamma}(\underline{R}) \rangle, \tag{4}$$

where $\underline{R} = \{\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_N\}$ denotes all nuclear coordinates.

These densities uniquely determine the external potentials through the generalized Hohenberg–Kohn theorem.The grand canonical potential is therefore expressed as

$$\Omega[n, \chi, \Gamma] = F[n, \chi, \Gamma] + \int d^3r \, n(\mathbf{r}) v^{e}_{\text{ext}}(\mathbf{r}) - \int d^3r d^3r' \chi(\mathbf{r}, \mathbf{r}') \Delta^{*}_{\text{ext}}(\mathbf{r}, \mathbf{r}') + \int d^3R \, \Gamma(\underline{R}) v^{n}_{\text{ext}}(\underline{R}). \tag{5}$$

Here $F[n, \chi, \Gamma]$ is the universal free-energy functional, v^{e}_{ext} is the external electronic potential, Δ_{ext} is the external pairing field, and v^{n}_{ext} is the external nuclear potential.

The interacting system is mapped to an auxiliary Kohn-Sham system.

The electronic subsystem satisfies the Kohn–Sham Bogoliubov–de Gennes equations

$$\left[-\frac{\nabla^2}{2} + v^{e}_{s}(\mathbf{r}) - \mu \right] u_i(\mathbf{r}) + \int d^3r' \Delta_s(\mathbf{r}, \mathbf{r}') v_i(\mathbf{r}') = E_i u_i(\mathbf{r}), \tag{6}$$

$$- \left[-\frac{\nabla^2}{2} + v^{e}_{s}(\mathbf{r}) - \mu \right] v_i(\mathbf{r}) + \int d^3r' \Delta^{*}_{s}(\mathbf{r}, \mathbf{r}') u_i(\mathbf{r}') = E_i v_i(\mathbf{r}). \tag{7}$$

Here:

- $u_i(\mathbf{r})$ and $v_i(\mathbf{r})$ are quasiparticle particle and hole amplitudes,
- E_i are quasiparticle excitation energies,
- $v^{e}_{s}(\mathbf{r})$ is the effective Kohn–Sham electronic potential,
- μ is the chemical potential,
- $\Delta_s(\mathbf{r}, \mathbf{r}')$ is the effective superconducting pairing potential.

The quasiparticle energies are

$$E_i = \sqrt{\xi_i^2 + |\Delta_i|^2}, \tag{8}$$

where $\xi_i = \epsilon_i - \mu$, with ϵ_i being the normal-state Kohn–Sham eigenvalue.

The electron density is calculated self-consistently as

$$n(\mathbf{r}) = \sum_{i} \left[1 - \frac{\xi_i}{E_i} \tanh\left(\frac{\beta E_i}{2} \right) \right] |\phi_i(\mathbf{r})|^2, \tag{9}$$

where $\phi_i(\mathbf{r})$ are normal-state wavefunctions and $\beta = 1/k_B T$ is the inverse temperature.

The anomalous density is

$$\chi(\mathbf{r}, \mathbf{r}') = \frac{1}{2} \sum_{i} \frac{\Delta_i}{E_i} \tanh\left(\frac{\beta E_i}{2} \right) \phi_i(\mathbf{r}) \phi^{*}_{i}(\mathbf{r}'). \tag{10}$$

This equation explicitly shows how Cooper pairing depends on the superconducting gap Δ_i.

The nuclear subsystem satisfies

$$\left[-\sum_{\alpha} \frac{\nabla^2_{\alpha}}{2M_{\alpha}} + v^{n}_{s}(\underline{R}) \right] \Phi_l(\underline{R}) = E_l \Phi_l(\underline{R}), \tag{11}$$

where M_α is the mass of nucleus α, v^{n}_{s} is the nuclear Kohn–Sham potential, and Φ_l are nuclear wavefunctions.

Expanding around equilibrium positions yields phonon modes described by

$$\hat{H}_{ph} = \sum_{\lambda, q} \Omega_{\lambda q} \left(\hat{b}^{\dagger}_{\lambda q} \hat{b}_{\lambda q} + \frac{1}{2} \right), \tag{12}$$

where $\Omega_{\lambda q}$ is the phonon frequency for branch λ and wavevector q, while $\hat{b}^{\dagger}_{\lambda q}$ and $\hat{b}_{\lambda q}$ are phonon creation and annihilation operators.

The superconducting gap equation is

$$\Delta_i = \Delta_{Hxc,i}, \tag{13}$$

where $\Delta_{Hxc,i}$ is the Hartree-exchange-correlation pairing potential.

Near the superconducting transition temperature,it becomes

$$\tilde{\Delta}_i = -\frac{1}{2}\sum_j \mathcal{F}_{Hxc,ij}\frac{\tanh(\beta\xi_j/2)}{\xi_j}\tilde{\Delta}_j, \tag{14}$$

where $\mathcal{F}_{Hxc,ij}$ is the exchange-correlation kernel responsible for effective pairing interaction.

The practical implementation follows the self-consistent iterative algorithm illustrated in Fig. 1 of Ref. [1]. Starting from an initial guess for v_s^e, Δ_s,and v_s^n,the electronic and nuclear equations are solved to obtain updated densities.These densities generate new Kohn–Sham potentials through functional derivatives,and the process is repeated until convergence is achieved.

This self-consistent loop forms the computationl foundation of superconducting density functional theory and enables first-principles prediction of superconducting properties.

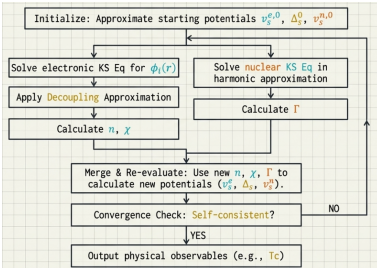


Fig. 1 The Self-Consistent Computational Engine

3 Results and Analysis

The central achievement of the superconducting density functional formalism is that it provides a fully *ab initio* description of superconductivity without introducing empirical parameters such as the Coulomb pseudopotential μ^* required in Eliashberg theory.Instead, both electron-electron repulsion and electron-phonon attraction are incorporated self-consistently through the exchange correlation functional.

The exchange-correlation free energy is expanded perturbatively as

$$F_{xc} = F_{xc}^{el} + F_{xc}^{ph} + F_{xc}^{mix}, \tag{15}$$

where F_{xc}^{el} represents purely electronic exchange contributions, F_{xc}^{ph} contains phonon-mediated pairing interactions,and F_{xc}^{mix} includes higher-order mixed corrections [1].

The purely electronic exchange contribution is

$$F_{xc}^{el} = -\frac{1}{4}\sum_{ij} v_{ij}\left(1 - \frac{\xi_i}{E_i}\tanh\frac{\beta E_i}{2}\right)\left(1 - \frac{\xi_j}{E_j}\tanh\frac{\beta E_j}{2}\right), \tag{16}$$

where v_{ij} are screened Coulomb matrix elements.This term produces the repulsive interaction that suppresses superconducting pairing.

The phonon-mediated contribution appears at second order and is written as

$$F_{xc}^{ph} = \frac{1}{2}\sum_{\lambda,q}\sum_{ij}|g_{ij}^{\lambda q}|^2\frac{\Delta_i\Delta_j^*}{E_iE_j}I(E_i,E_j,\Omega_{\lambda q}), \tag{17}$$

where $g_{ij}^{\lambda q}$ are electron-phonon coupling matrix elements, $\Omega_{\lambda q}$ are phonon frequencies, and $I(E_i,E_j,\Omega)$ contains retardation effects.

The electron-phonon interaction strength is characterized by the Eliashberg spectral function

$$\alpha^2F(\Omega) = \frac{1}{N(0)}\sum_{ij,\lambda,q}|g_{ij}^{\lambda q}|^2\delta(\xi_i)\delta(\xi_j)\delta(\Omega-\Omega_{\lambda q}), \tag{18}$$

where $N(0)$ is the electronic density of states at the Fermi level.

The corresponding dimensionless coupling constant is

$$\lambda = 2\int d\Omega\frac{\alpha^2F(\Omega)}{\Omega}. \tag{19}$$

This parameter determines the strength of phonon-mediated superconducting attraction.

As shown in Fig. 2, the phonon-mediated pairing kernel is strongly peaked near the Fermi surface. This confirms that superconducting pairing arises predominantly from low-energy electronic excitations, consistent with conventional BCS and Eliashberg theory.

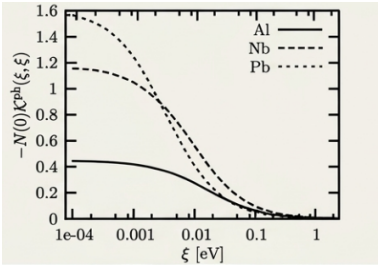


Fig. 2 Energy dependence of the phonon-mediated pairing kernel for Al,Nb,and Pb.The sharp enhancement near the Fermi surface shows that superconducting pairing is dominated by low-energy electronic states.

Near the superconducting transition temperature,the nonlinear gap equation reduces to the linearized form

$$\Delta_i = -\frac{1}{2}\sum_j K_{ij}\frac{\tanh(\beta\xi_j/2)}{\xi_j}\Delta_j, \tag{20}$$

where the total pairing kernel is

$$K_{ij} = K_{ij}^{ph} + K_{ij}^{el}. \tag{21}$$

Here K_{ij}^{ph} provides attractive phonon-mediated interaction, while K_{ij}^{el} represents repulsive electronic Coulomb interaction. Superconductivity occurs when the attractive contribution dominates.

For a simplified BCS-like approximation with constant kernel and renormalization factor Z, the critical temperature becomes

$$T_c \sim \exp\left(-\frac{1+Z(0)}{N(0)K(0)}\right). \tag{22}$$

This has the same structure as McMillan's empirical formula

$$T_c = \frac{\omega_{ln}}{1.20}\exp\left[-\frac{1.04(1+\lambda)}{\lambda-\mu^*(1+0.62\lambda)}\right]. \tag{23}$$

Comparing these relations gives the correspondence

$$N(0)K(0) \approx -\lambda, \qquad Z(0) \approx \lambda. \tag{24}$$

This demonstrates that the density functional formalism naturally reproduces the strong-coupling behavior of Eliashberg theory while eliminating the empirical parameter μ^*.

A major advantage of this approach is predictive capability. Once the electronic band structure and phonon spectrum are obtained from first-principles calculations, superconducting quantities such as the gap function and critical temperature can be determined directly.

Practical implementations still require approximations such as the decoupling approximation and the neglect of higher-order vertex corrections through Migdal's theorem. These approximations work well for conventional phonon-mediated superconductors such as aluminum, niobium, and lead, but may be insufficient for strongly correlated unconventional superconductors.

Despite these limitations, the theory successfully unifies density functional theory, BCS pairing, and Eliashberg strong-coupling physics into a single self-consistent first-principles framework for superconductivity.

4 Conclusion

The superconducting density functional theory developed by Lüders *et al.* represents a major advancement in the theoretical description of superconducting materials. By extending conventional density functional theory to include the anomalous density and nuclear degrees of freedom, the formalism provides a rigorous first-principles framework for describing superconductivity without relying on empirical fitting parameters.

A key achievement of this theory is the unified treatment of electron-electron and electron-phonon interactions within the exchange-correlation functional.This allows superconducting pairing, retardation effects,and strong-coupling corrections to emerge naturally from the self-consistent solution of the Kohn–Sham equations.The resulting formalism reproduces the essential physics of both BCS and Eliashberg theory while maintaining the predictive power of *ab initio* electronic structure methods.

The analysis of the linearized gap equation demonstrates that the theory naturally recovers McMillan-like behavior for the superconducting critical temperature, confirming its consistency with established strong-coupling superconductivity theory. At the same time, the absence of adjustable parameters makes the approach significantly more predictive than conventional Eliashberg calculations.

Although practical implementations require approximations such as the decoupling approximation and the neglect of higher-order vertex corrections,the formalism has been shown to accurately describe conventional phonon-mediated superconductors such as aluminum,niobium,and lead. This confirms its effectiveness as a quantitative computational tool.

Overall, superconducting density functional theory establishes an important bridge between quantum many-body theory and modern computational materials science. It provides a powerful foundtion for predicting superconducting properties directly from atomic structure and opens new possibilities for the theoretical discovery and design of novel superconducting materials.

5 Bibliography

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